

CS383 - Software Engineering

Software Design (behavioral modeling 2)

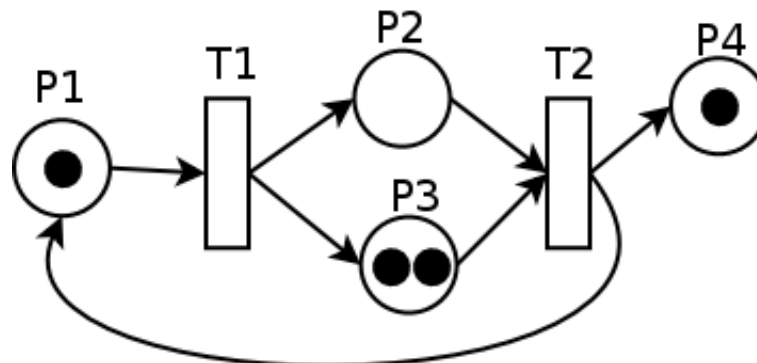
Event-driven models



- Describes how the system and its components react to internal and/or external triggered events
- A system has a finite number of states
- Moving from state to another requires an event
- Appropriate for real-time systems
- A state of a system is not reached until a certain state(s) is/are finished

Event-driven models

- Models used for modeling events are
 - UML state / state machine diagram (software and system engineering)
 - Petri net, finite-state automata: mathematical modelling of system states and execution
 - Prism: multi-objective system modeling and simulation



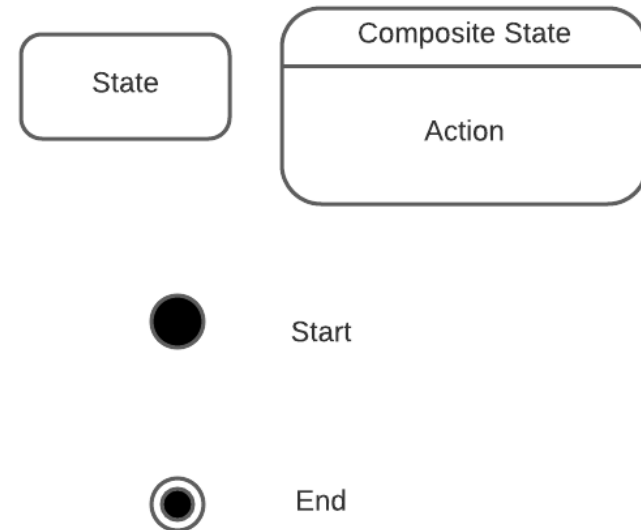
State diagram



- At any given state, only some of all possible events & actions are appropriate
 - E.g. if light is off then can turn it on but can't turn it off
- Often, when the state changes the appropriate events & actions change
 - E.g. after turning light on can turn it off but can't turn it on
- Important to constrain available events & actions to prevent
 - action sequences which might damage system (and user)
 - e.g. can't open washing machine door during wash cycle
 - user confusion, e.g. can't close file which is already closed

State diagram (cont.)

- A state describes the condition of an object at a given point in its lifecycle
 - Used in early stages of development to know the lifecycle of an object
- It describes all possible states of an object & the paths from one state to another
- Few symbols are shared with activity diagram
- State in state diagram can have multiple incoming and outgoing arrows
- For multiple objects involvement, it is better to use sequence diagrams (interaction models)

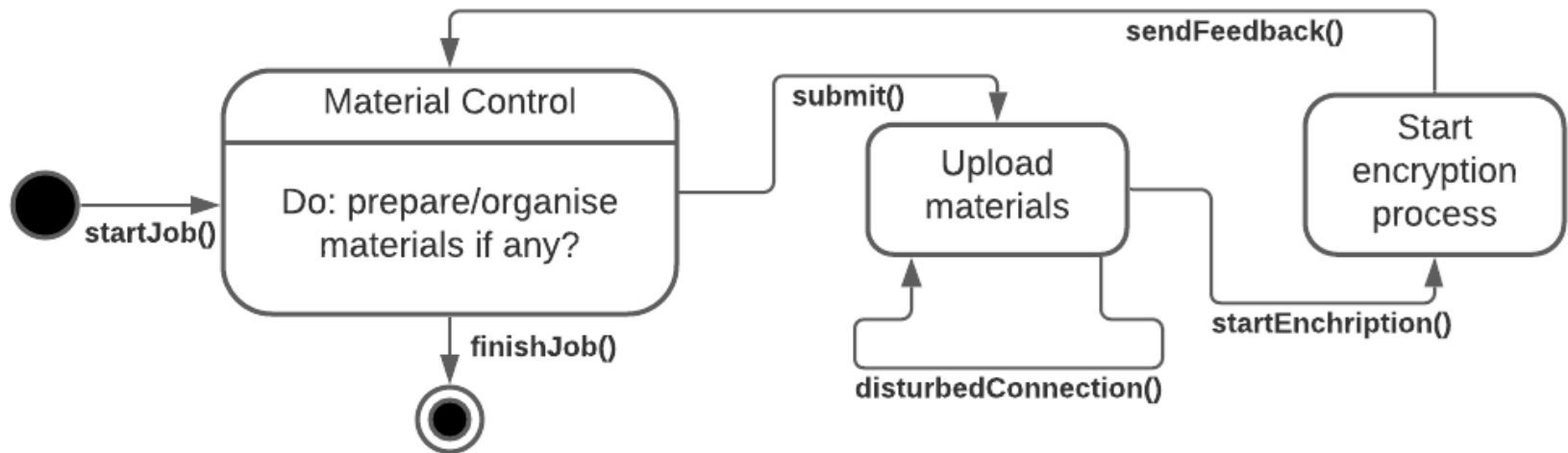


State diagram (cont.)

- Between two states, there is a transition
 - Basically, it is a defined action that is carried out on an object of a state
 - Action must take place when only conditions and events are satisfied in the source state
 - An event is an occurrence that triggers a transition
 - Transition can have also condition or guard. Use [...] for that.
 - Eg. in the figure shown, all transitions are labeled with an event flip
 - Two of which are self-transitions



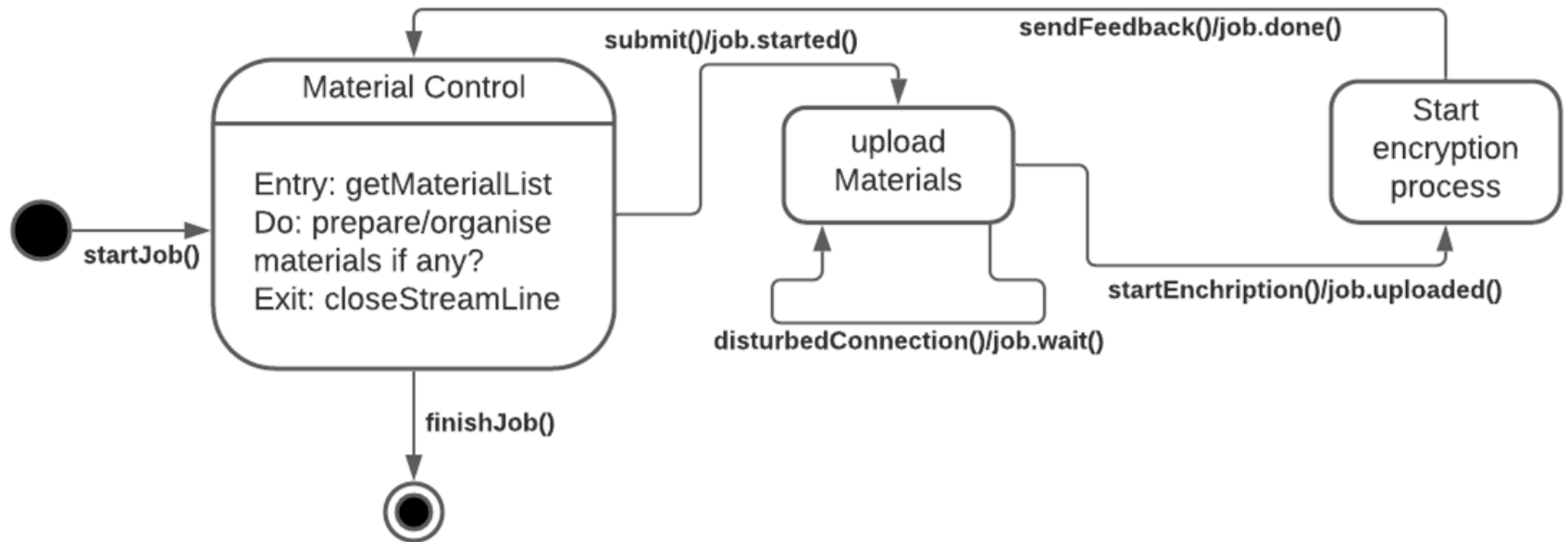
State diagram examples



State diagram (cont.)

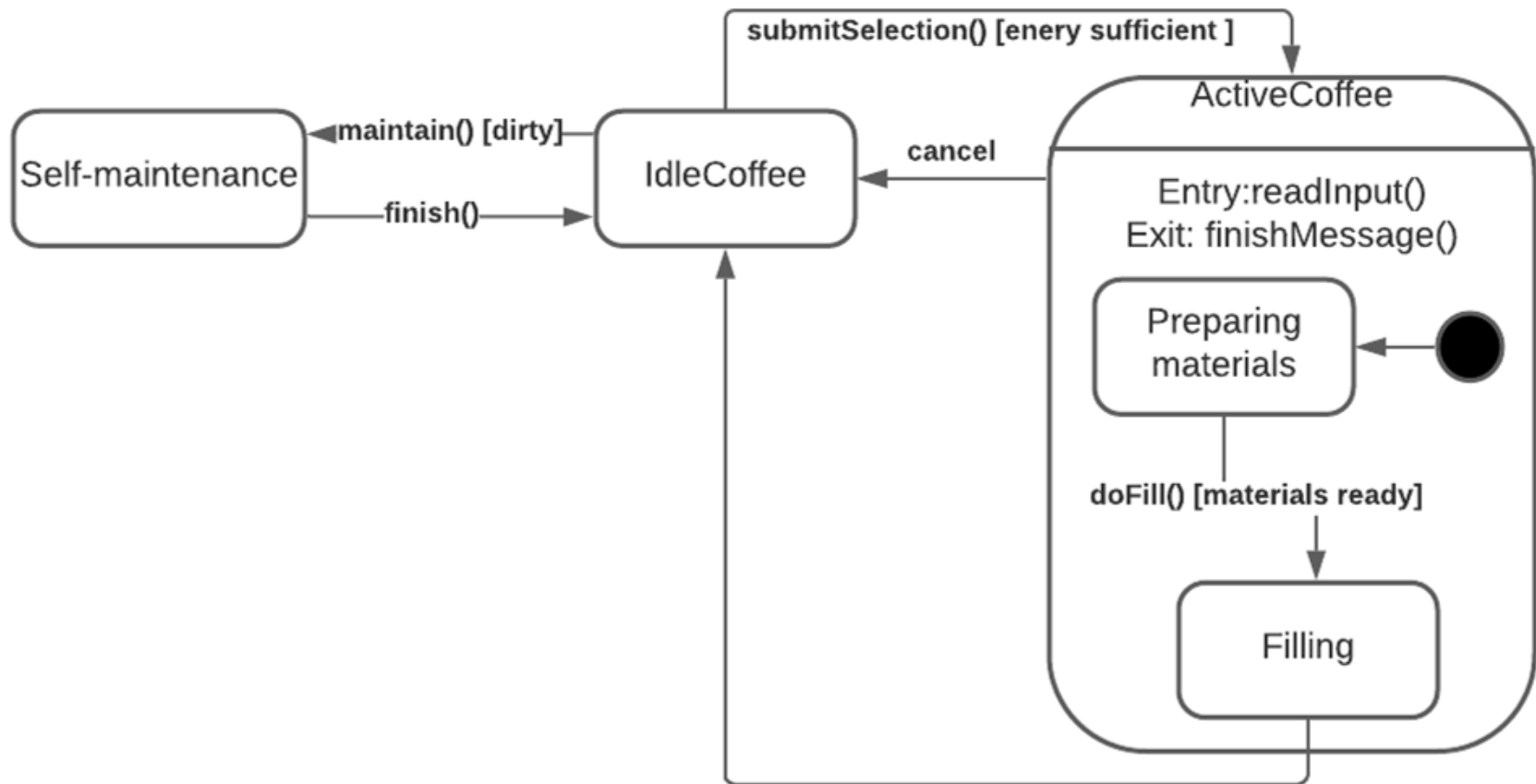
- Actions can be provided in state diagram
 - Transitions
 - After transition call use / and call whatever action in target object
 - States
 - Using composite states use three actions
 - Entry: action to do on entry of the state
 - Do: action(s) after entry and while in state
 - Exit: action to do on leaving the state
 - A single state use only one action (decomposition)

State diagram examples (cont.)



Nested states

- Modeled using composite state



How to design/model a system



- Brainstorm all states of the system
- Determine what are the single states and what are the composite states
- Analyse the events/actions for transitioning from one state to another
- Associate the necessary conditions/constraints for the events/actions

State diagram exercise 1



A university computer has two general states: idle and active. In the idle case, when the user activates the computer (press any button on keyboard), the computer activates. In active case, the user can press escape button anytime to exit. Also, the computer waits only 1 minutes for the user to enter his/her credentials for authorization. If authorized, the computer allows a session of 15 minutes for the user. When the session ends, the user is prompted immediately to renew the session for another 15 minutes. If the session ends without renewal request, the computer goes back to the idle state.

State diagram exercise 2



Provide a state diagram that models a centralized air system according to the following actions:

1. Fan model: triggered whenever the user presses the turn on while the system is in standby mode or called by Cooling and Heating modes
2. Cooling mode: starts when the room temperature is above the room temperature set up by customer. Once the temperature reaches or goes below the user temperature the system goes to fan mode.
3. Heating mode: starts when the room temperature is less than the user temperature. Once the temperature reaches or goes above the user temperature the system goes to fan mode.
4. Standby mode: initial state of the system, triggered whenever the user presses the turn off button from any state.

State diagram exercise 3



Provide a state diagram that models an oven with the following modes:

1. Standby mode: this is triggered whenever the user presses the turn-on button. Also, this mode allows the user to enter the operation time and the target stoves to be working. If the user presses the off button, the state of the oven goes to Off mode.
2. Heating mode: this starts whenever the user presses the execute button in the standby mode. This mode is not triggered or activated unless the user has entered the heating program and has placed an item with weight (at least 100 grams) on top of the target stove. If this mode is activated, the program is loaded, and the target stove starts heating. After the operation time finishes, the oven goes to standby mode.
3. Off mode: this mode is activated whenever the user presses the turn-off button.

State diagram exercise 4



Provide a state diagram that model the states of a pedestrian's (walking people) traffic light with the following modes:

1. EmergencyMode: this mode (which is also the initial mode) is activated whenever a malfunction (a hardware or software fault) appears in the RedMode or GreenMode. In this mode, only the amber (yellowish) color is turned ON and OFF constantly each second. If the traffic light is fixed, it goes into the RedMode.
2. RedMode: pedestrians who wish to cross a road safely can press the STOP button to submit a request to stop the road traffic of vehicles. If the button is pressed, the pedestrian traffic light goes into the GreenMode after 10 seconds of pressing. Otherwise, only the red color is turned ON. In case of a malfunction, the traffic light goes into the EmergencyMode.
3. GreenMode: the pedestrian traffic light only shows green color firstly for 7 seconds, and then the traffic light goes back into the RedMode. Also, the pedestrian traffic light produces a light beep sound during the 7 seconds. In case of a malfunction, the traffic light goes into the EmergencyMode.